

- 3 (a) (i) mean kinetic energy M1
of the atoms / molecules / particles A1
(ii) at absolute zero, atoms have no kinetic energy B1 [3]
- (b) (i) $pV = nRT$ C1
 $n = (1.2 \times 10^5 \times 2.0 \times 10^{-2}) / (8.31 \times 310)$ C1
 $= 0.93 \text{ mol}$ A1
(ii) total amount $= (1.20 + 0.93)$ C1
 $(1.20 + 0.93) = (4.0 \times 10^{-2} \times p) / (8.31 \times 310)$ C1
 $p = 1.37 \times 10^5 \text{ Pa}$ A1 [6]